# Fahim Tajwar

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## EDUCATION

**Carnegie Mellon University** **Pittsburgh, PA**

Doctor of Philosophy (PhD), Machine Learning 2023 – Current

- Advisor: Ruslan Salakhutdinov & Jeff Schneider
- Research Focus: Large Language Models (LLMs), Reinforcement Learning, Multi-turn Interactions, Decision Making

**Stanford University** **Stanford, CA**

Master of Science (MS), Computer Science (AI/ML) 2023

Bachelor of Science (BS) with Distinction, Mathematics 2022

- Research Advisor: Chelsea Finn
- Research Focus: Distribution Shift Robustness, Out-of-Distribution (OOD) Detection, Selective Classification

## INDUSTRY EXPERIENCE

**Research Scientist Intern, GenAI, Meta Platforms** May 2025 – August 2025

**Software Engineer Intern, Meta Platforms** June 2022 – September 2022

**Software Engineer Intern, Cadence Design Systems** June 2020 – September 2020

## PREPRINTS (* Equal Contribution)

Maximum Likelihood Reinforcement Learning 2026

Fahim Tajwar*, Guanning Zeng*, Yueer Zhou, Yuda Song, Daman Arora, Yiding Jiang, Jeff Schneider, Ruslan Salakhutdinov, Haiwen Feng, Andrea Zanette

Preprint, 2026

Can Large Reasoning Models Self-Train? 2025

Sheikh Shafayat*, Fahim Tajwar*, Ruslan Salakhutdinov, Jeff Schneider, Andrea Zanette

Preprint, 2025

## PUBLICATIONS (* Equal Contribution)

Accelerating Diffusion Planners in Offline RL via Reward-Aware Consistency Trajectory Distillation 2026

Xintong Duan*, Yutong He*, Fahim Tajwar, Ruslan Salakhutdinov, J Zico Kolter, Jeff Schneider

International Conference on Learning Representations (ICLR), 2026

State Combinatorial Generalization in Decision Making with Conditional Diffusion Models 2025

Xintong Duan, Yutong He, Fahim Tajwar, Wen-Tse Chen, Ruslan Salakhutdinov, Jeff Schneider

Transactions on Machine Learning Research (TMLR), 2025

Reasoning as an Adaptive Defense for Safety 2025

Taeyoun Kim, Fahim Tajwar, Aditi Raghunathan, Aviral Kumar

Conference on Neural Information Processing Systems (NeurIPS), 2025

Retrospective In-Context Learning for Temporal Credit Assignment with Large Language Models 2025

Wen-Tse Chen, Jiayu Chen, Fahim Tajwar, Hao Zhu, Xintong Duan, Ruslan Salakhutdinov, Jeff Schneider

Conference on Neural Information Processing Systems (NeurIPS), 2025

Training a Generally Curious Agent 2025

Fahim Tajwar*, Yiding Jiang*, Abitha Thankaraj, Sumaita Sadia Rahman, J Zico Kolter, Jeff Schneider, Ruslan Salakhutdinov

International Conference on Machine Learning (ICML), 2025 **(Oral)**

Preference Fine-Tuning of LLMs Should Leverage Suboptimal, On-Policy Data 2024
Fahim Tajwar*, Anikait Singh*, Archit Sharma, Rafael Rafailov, Jeff Schneider, Tengyang Xie, Stefano Ermon, Chelsea Finn, Aviral Kumar
International Conference on Machine Learning (ICML), 2024

Conservative Prediction via Data-Driven Confidence Minimization 2024
Caroline Choi*, Fahim Tajwar*, Yoonho Lee*, Huaxiu Yao, Ananya Kumar, Chelsea Finn
Transactions on Machine Learning Research (TMLR), 2024

Surgical Fine-Tuning Improves Adaptation to Distribution Shifts 2023
Yoonho Lee*, Annie S. Chen*, Fahim Tajwar, Ananya Kumar, Huaxiu Yao, Percy Liang, Chelsea Finn
International Conference on Learning Representations (ICLR), 2023

When to Ask for Help: Proactive Interventions in Autonomous Reinforcement Learning 2022
Annie Xie*, Fahim Tajwar*, Archit Sharma*, Chelsea Finn
Conference on Neural Information Processing Systems (NeurIPS), 2022

Do Deep Networks Transfer Invariances Across Classes? 2022
Allan Zhou*, Fahim Tajwar*, Alexander Robey, Tom Knowles, George J. Pappas, Hamed Hassani, Chelsea Finn
International Conference on Learning Representations (ICLR), 2022

Scalable Deep Learning to Identify Brick Kilns and Aid Regulatory Capacity 2021
Jihyeon Lee*, Nina R. Brooks*, Fahim Tajwar, Marshall Burke, Stefano Ermon, David B. Lobell, Debashish Biswas, Stephen P. Luby
Proceedings of the National Academy of Sciences (PNAS), 2021

## TECHNICAL REPORTS (* Equal Contribution)

Self-Regulation and Requesting Interventions 2025
So Yeon Min, Yue Wu, Jimin Sun, Max Kaufmann, Fahim Tajwar, Yonatan Bisk, Ruslan Salakhutdinov
ArXiv, 2025

Offline Retraining for Online RL: Decoupled Policy Learning to Mitigate Exploration Bias 2023
Max Sobol Mark*, Archit Sharma*, Fahim Tajwar, Rafael Rafailov, Sergey Levine, Chelsea Finn
ArXiv, 2023

No True State-of-the-Art? OOD Detection Methods are Inconsistent across Datasets 2021
Fahim Tajwar, Ananya Kumar*, Sang Michael Xie*, Percy Liang
Workshop on Uncertainty and Robustness in Deep Learning (UDL) @ ICML 2021

## AWARDS

MIT Presidential Fellowship (Declined) 2023
Top Reviewer, Conference on Neural Information Processing Systems (NeurIPS) 2023
University Distinction, top 15% of the graduating class, Stanford University 2022
Tau Beta Pi Engineering Honor Society 2020
Bronze Medal, 48th International Physics Olympiad (**IPhO**), Indonesia 2017
Bronze Medal, 47th International Physics Olympiad (**IPhO**), Switzerland Liechtenstein 2016

## INVITED TALKS

- Invited talk at BRAC University 2025
- Guest lecture at Korea Advanced Institute of Science & Technology (KAIST) 2024
- Invited talk at ServiceNow 2024